

SANDY BROWN

Consultants in Acoustics, Noise & Vibration

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Greenwich Millennium Village, Phases 3, 4 & 5 Building 304

Building 304 facade sound insulation mark ups – balconies and windows

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1 Building 304 facade sound insulation mark ups – balconies and windows

The following pages contain marked up floor plans for building 304 demonstrating how the design principles for opening windows and partially enclosed balconies can be applied. The minimum sound insulation performances for doors and windows on the various facades of the building are also provided.

The mark ups are based on the latest internal layouts issued by Jestico + Whiles for the building.

Table 1 provides a key to the tables for each facade mark-up in the following pages of this document.

Table 1 Key to facade mark-up table

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz} *)		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
XX	XX	XX	XX	XX	XX	XX	XX	XX	XX%	XX%	Yes / No	Yes / No	XX	XX	

Window / door reference number

Criterion 1: *L*_{Aeq} ≥ 45 dB - window fixed closed unless adequately shrouded

Criterion 2: *L*_{A63Hz} ≥ 40 dB - window recommended to be closed unless shrouded; predicted levels compared to Acoustic Report to determine required *D*_{63Hz} for window / door

Criteria 3 & 4: predicted levels which are compared to Acoustic Report to determine required window / door sound insulation performance

* This number is approximated from the broadband models of dredger noise, which dominates at 63 Hz. The actual level at 63 Hz is either equal to this number or as much as 3 dB lower. See discussion in cover report (17470-R02-C GMV3 Buildings 302, 303 & 304 -Internal and facade levels- acoustic Cond 87 Report, dated 17 April 2018).

All levels given are A-weighted facade noise levels in dB.

- The first level in each column (AO) is the noise level predicted with just the buildings existing ‘At Occupation’ of Buildings 304, namely Buildings 101 to 109, 201, 204 to 210 and 504.
- The second level in each column (MP) is the noise levels predicted with all the buildings of the masterplan in place – including all later parcels.

The percentages of allowable open areas are based on balcony/ shroud’s frontage only. The ends, top and floor may need to be solid, as noted herein.

Where no balcony / shroud is present, the percentage of allowable open area and the decision on whether a window may open will require further measures as noted herein. The additional screening required will be designed so that the openings are shielded from the wharf / dredger noise to ensure the internal noise levels can still be met. Please refer to the architectural drawings for more details on this.

The window / door sound insulation values given (ie, *D*_w + *C*_{tr} and *D*_{63Hz}) assume the window / door is more than 50% of the facade of the room and the remaining solid portion has at least the same performance. Where windows are a small portion of the facade and the surrounding solid part of the facade is at least 5 dB better than the windows, a slight relaxation in the requirement for the window / door as shown in Table 2 is then possible.

Table 2 Composite facade sound insulation performance, solid v glazed

Sound insulation required (from Figures on the following pages)	Minimum sound insulation of solid portion	Minimum sound insulation of fenestration if less than 20%/30%/50% of room facade
<i>D</i> _w + <i>C</i> _{tr} ≥ 30 dB	<i>D</i> _w + <i>C</i> _{tr} ≥ 35 dB	<i>D</i> _w + <i>C</i> _{tr} ≥ 27/28/29 dB
<i>D</i> _w + <i>C</i> _{tr} ≥ 25 dB	<i>D</i> _w + <i>C</i> _{tr} ≥ 30 dB	<i>D</i> _w + <i>C</i> _{tr} ≥ 22/23/24 dB
<i>D</i> _w + <i>C</i> _{tr} ≥ 20 dB	<i>D</i> _w + <i>C</i> _{tr} ≥ 25 dB	<i>D</i> _w + <i>C</i> _{tr} ≥ 17/18/19 dB

2 Building 304 Block J– Ground level

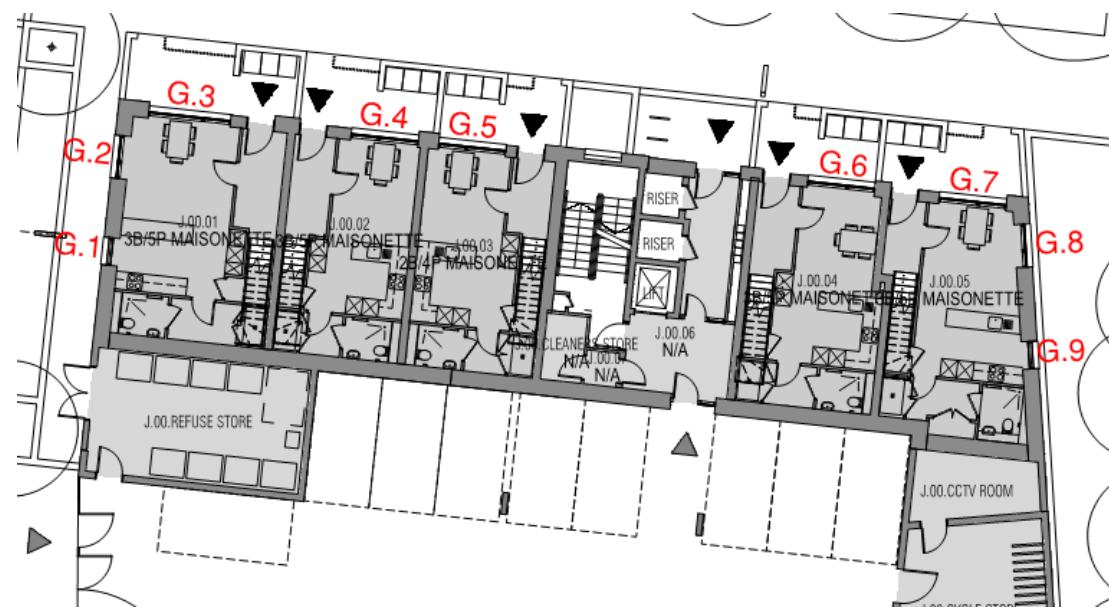


Figure 1 Building 304 Block J– Ground level floor plan

Table 3 Building 304 Block J – Ground level predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
G.1	41	41	31	34	60	60	61	61	100%	100%	YES	YES	31	15	
G.2	41	41	31	34	60	60	61	61	100%	100%	YES	YES	31	15	
G.3	41	41	34	34	60	60	61	61	100%	100%	YES	YES	31	15	
G.4	41	42	34	37	60	60	61	61	100%	100%	YES	YES	31	15	
G.5	41	41	37	37	61	60	61	61	100%	100%	YES	YES	32	15	
G.6	41	42	37	37	61	61	63	62	100%	100%	YES	YES	32	15	
G.7	41	42	37	35	62	61	62	63	100%	100%	YES	YES	33	15	
G.8	41	42	33	33	65	65	67	67	100%	100%	YES	YES	36	15	
G.9	41	42	35	35	65	65	67	67	100%	100%	YES	YES	36	15	

3 Building 304 Block K– Ground level

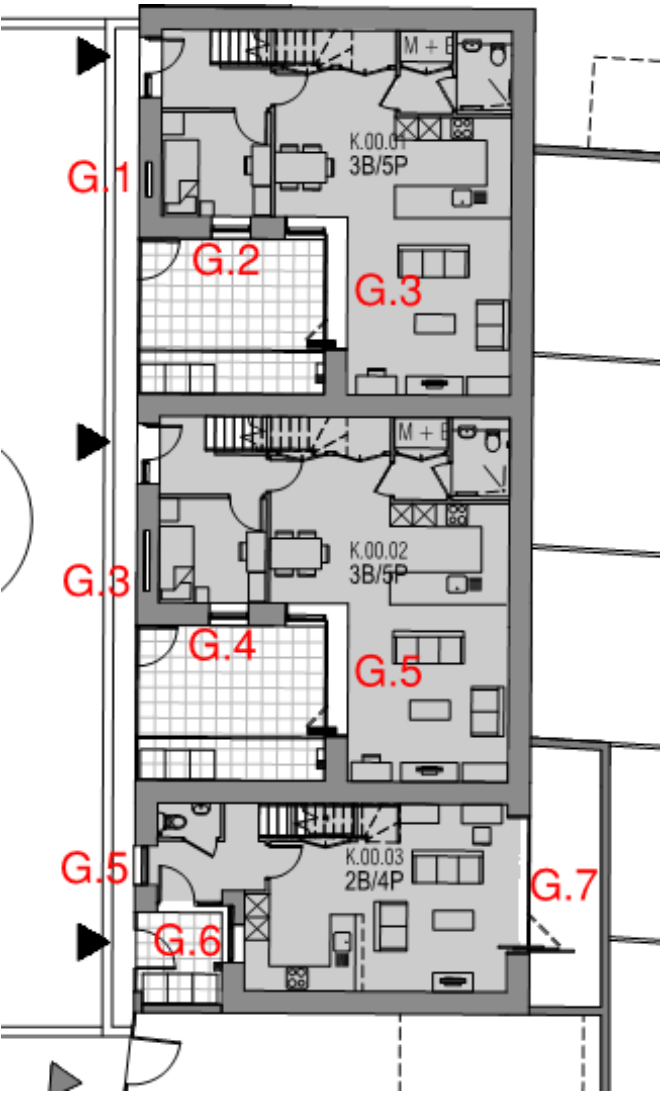


Figure 2 Building 304 Block K– Ground level floor plan

Table 4 Building 304 Block K – Ground level predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
G.1	39	39	31	31	47	45	53	51	100%	100%	YES	YES	18	15	
G.2	41	41	32	32	46	44	55	49	100%	100%	YES	YES	17	15	
G.3	41	41	32	32	46	44	55	49	100%	100%	YES	YES	17	15	
G.4	41	41	32	33	48	44	56	49	100%	100%	YES	YES	19	15	
G.5	41	41	33	33	49	44	55	50	100%	100%	YES	YES	18	15	
G.6	41	41	35	35	49	44	54	50	100%	100%	YES	YES	18	15	
G.7	41	41	36	36	48	48	54	53	100%	100%	YES	YES	19	15	

4 Building 304 Block L– Ground level

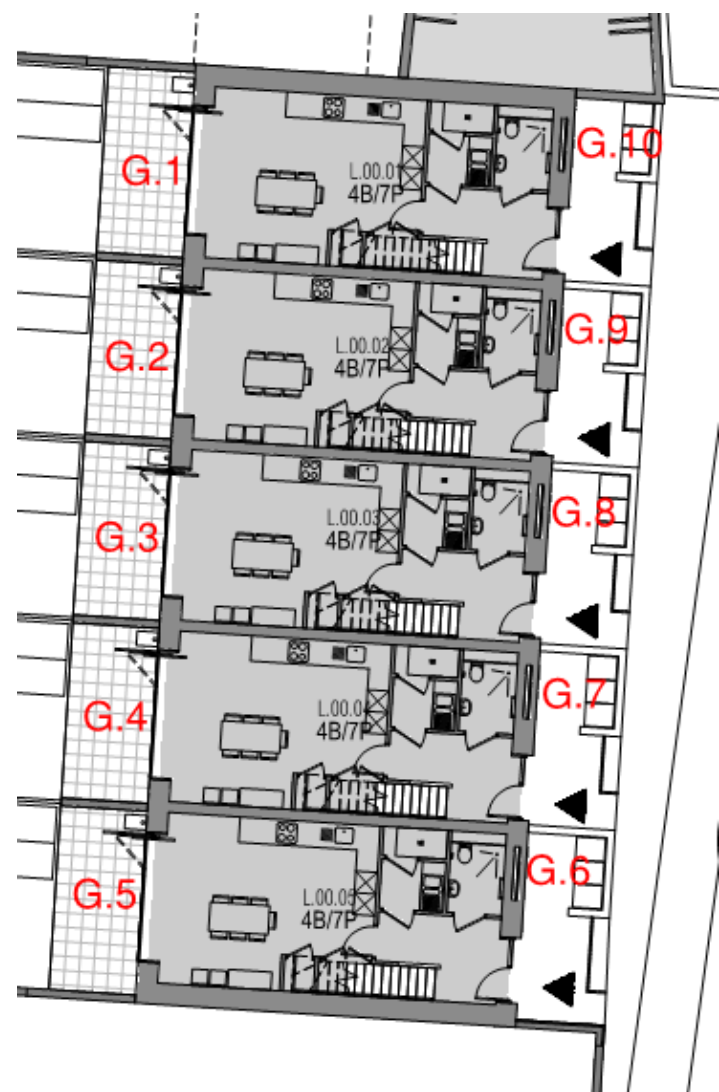


Figure 3 Building 304 Block L– Ground level floor plan

Table 5 Building 304 Block L – Ground level predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (L_{Aeq})		Dredgers (L_{A63Hz})		All sources night (L_{Aeq})		All sources day (L_{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	D_w+C_{tr}	D_{63Hz}	
G.1	37	37	30	31	48	48	53	53	100%	100%	YES	YES	19	15	
G.2	37	37	31	31	46	56	51	51	100%	100%	YES	YES	27	15	
G.3	37	38	31	31	46	45	50	50	100%	100%	YES	YES	17	15	
G.4	38	38	31	32	46	45	50	50	100%	100%	YES	YES	17	15	
G.5	38	38	32	32	45	45	51	50	100%	100%	YES	YES	16	15	
G.6	42	42	37	36	65	65	67	67	100%	100%	YES	YES	36	15	
G.7	42	43	37	36	65	65	67	67	100%	100%	YES	YES	36	15	
G.8	42	43	36	35	65	65	67	67	100%	100%	YES	YES	36	15	
G.9	42	43	37	35	65	65	67	67	100%	100%	YES	YES	36	15	
G.10	42	42	36	35	65	65	67	67	100%	100%	YES	YES	36	15	

5 Building 304 Block M– Ground level

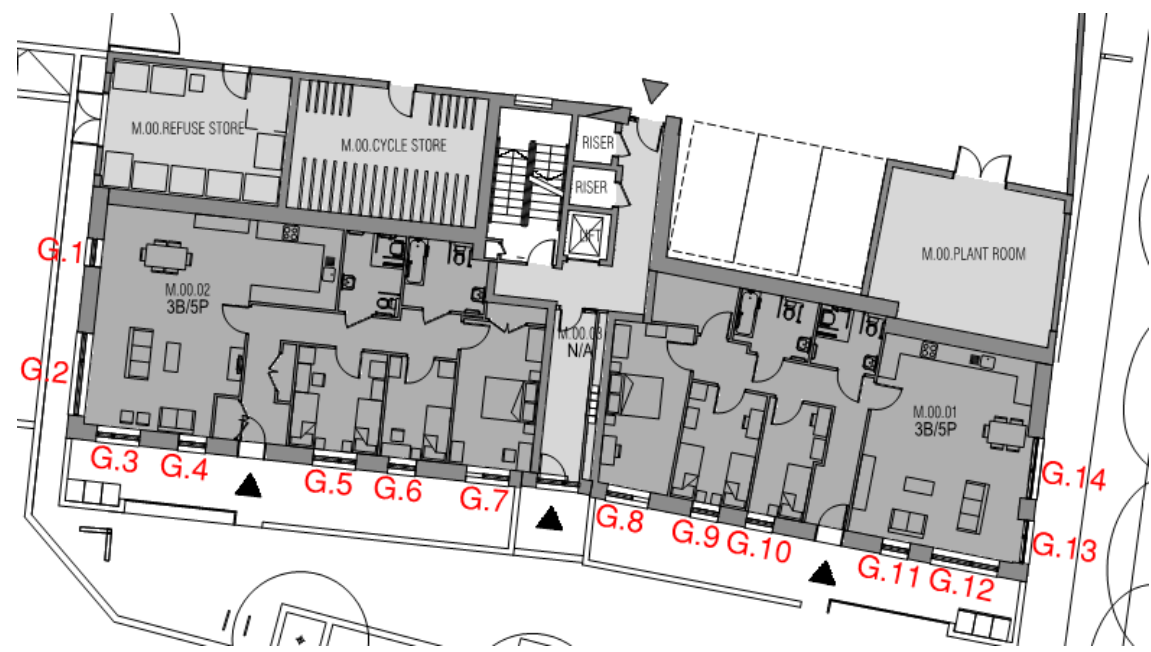


Figure 4 Building 304 Block M– Ground level floor plan

Table 6 Building 304 Block M – Ground level predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
G.1	42	41	27	27	51	46	57	53	100%	100%	YES	YES	22	15	
G.2	42	40	27	27	52	46	63	62	100%	100%	YES	YES	24	15	
G.3	40	41	27	27	55	51	63	60	100%	100%	YES	YES	26	15	
G.4	38	40	27	34	56	52	63	60	100%	100%	YES	YES	29	15	
G.5	38	40	27	34	56	53	63	61	100%	100%	YES	YES	29	15	
G.6	38	40	27	36	57	53	63	61	100%	100%	YES	YES	29	15	
G.7	38	40	27	36	57	53	63	61	100%	100%	YES	YES	29	15	
G.8	39	40	27	36	58	56	63	61	100%	100%	YES	YES	29	15	
G.9	39	40	27	36	59	57	64	62	100%	100%	YES	YES	30	15	
G.10	39	40	27	36	60	58	64	62	100%	100%	YES	YES	31	15	
G.11	41	40	27	35	61	61	65	64	100%	100%	YES	YES	32	15	
G.12	41	40	27	35	61	61	65	64	100%	100%	YES	YES	32	15	
G.13	42	42	36	36	65	65	67	67	100%	100%	YES	YES	36	15	
G.14	41	41	34	36	65	65	67	67	100%	100%	YES	YES	36	15	

6 Building 304 Block J– Level 1



Figure 5 Building 304 Block J– Level 1 floor plan

Table 7 Building 304 Block J – Level 1 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
1.1	42	42	35	35	61	61	61	61	100%	100%	YES	YES	31	15	
1.2	42	42	35	35	61	61	61	61	100%	100%	YES	YES	31	15	
1.3	41	42	35	35	61	61	61	61	100%	100%	YES	YES	31	15	
1.4	41	42	35	35	61	61	61	61	100%	100%	YES	YES	31	15	
1.5	42	42	37	35	61	61	62	61	100%	100%	YES	YES	31	15	
1.6	42	43	38	38	61	61	62	62	100%	100%	YES	YES	32	15	
1.7	42	42	38	38	61	62	63	63	100%	100%	YES	YES	33	15	
1.8	43	42	38	38	62	62	64	64	100%	100%	YES	YES	33	15	
1.9	42	42	35	35	64	65	67	66	100%	100%	YES	YES	36	15	
1.10	43	43	36	36	65	65	67	67	100%	100%	YES	YES	31	15	
1.11	43	43	36	36	64	64	67	66	100%	100%	YES	YES	35	15	
1.12	40	39	32	32	61	61	64	64	100%	100%	YES	YES	32	15	
1.13	39	39	31	31	58	58	60	62	100%	100%	YES	YES	29	15	
1.14	39	39	31	31	57	57	59	60	100%	100%	YES	YES	28	15	
1.15	40	39	33	33	56	56	58	58	100%	100%	YES	YES	27	15	
1.16	40	40	34	34	54	56	57	57	100%	100%	YES	YES	27	15	
1.17	40	40	34	34	54	54	57	56	100%	100%	YES	YES	25	15	
1.18	40	40	34	34	50	49	56	55	100%	100%	YES	YES	21	15	
1.19	41	41	36	36	49	48	55	54	100%	100%	YES	YES	21	15	
1.20	41	41	36	36	49	48	55	53	100%	100%	YES	YES	20	15	
1.21	41	41	36	36	49	47	55	53	100%	100%	YES	YES	20	15	
1.22	40	40	33	33	54	53	56	54	100%	100%	YES	YES	25	15	
1.23	39	39	32	32	57	56	59	58	100%	100%	YES	YES	28	15	

7 Building 304 Block K– Level 1

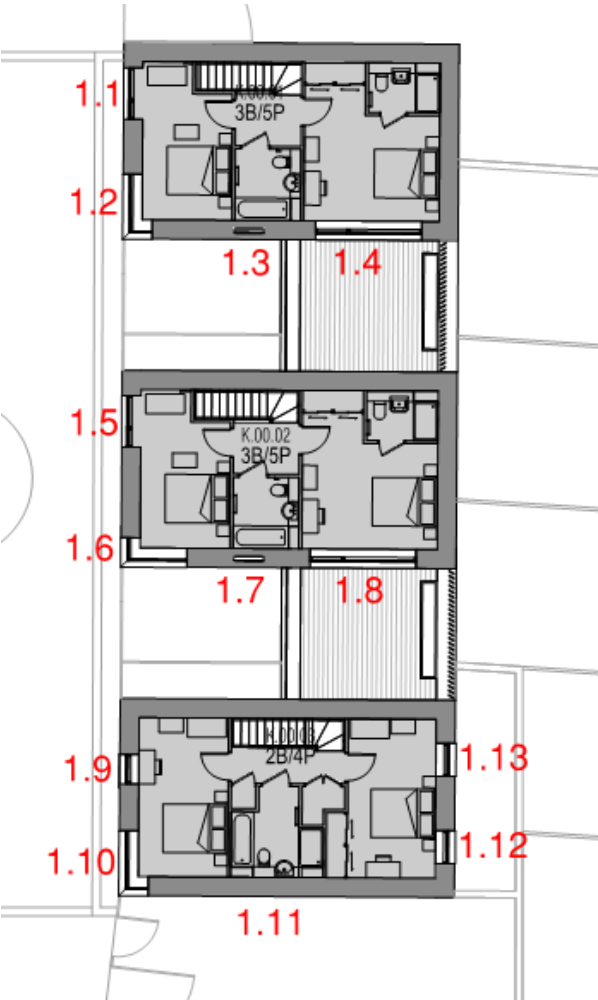


Figure 6 Building 304 Block K– Level 1 floor plan

Table 8 Building 304 Block K– Level 1 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
1.1	41	41	33	33	48	47	54	53	100%	100%	YES	YES	20	15	
1.2	42	42	33	34	48	46	54	52	100%	100%	YES	YES	20	15	
1.3	42	42	34	34	48	45	54	50	100%	100%	YES	YES	14	15	
1.4	42	40	34	34	48	45	53	51	100%	100%	YES	YES	19	15	
1.5	42	42	34	34	50	45	56	51	100%	100%	YES	YES	22	15	
1.6	42	42	36	36	50	45	55	50	100%	100%	YES	YES	21	15	
1.7	42	42	36	36	50	45	55	50	100%	100%	YES	YES	16	15	
1.8	42	41	36	36	50	45	53	52	100%	100%	YES	YES	21	15	
1.9	43	43	36	36	51	45	57	51	100%	100%	YES	YES	23	15	
1.10	43	43	36	36	51	45	57	51	100%	100%	YES	YES	23	15	
1.11	43	43	36	37	50	49	56	54	100%	100%	YES	YES	16	15	
1.12	42	42	35	37	48	49	56	54	100%	100%	YES	YES	22	15	
1.13	42	42	35	37	48	49	56	54	100%	100%	YES	YES	22	15	

8 Building 304 Block L– Level 1



Figure 7 Building 304 Block L – Level 1 floor plan

Table 9 Building 304 Block L – Level 1 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{Tr}	<i>D</i> _{63Hz}			
1.1	38	39	31	32	49	49	54	54	100%	100%	YES	YES	20	15	
1.2	38	38	32	32	48	47	52	51	100%	100%	YES	YES	19	15	
1.3	38	38	32	32	47	46	52	50	100%	100%	YES	YES	18	15	
1.4	38	38	32	32	47	46	52	50	100%	100%	YES	YES	18	15	
1.5	38	38	32	32	47	46	52	50	100%	100%	YES	YES	18	15	
1.6	38	38	34	34	47	45	51	50	100%	100%	YES	YES	18	15	
1.7	39	39	34	34	47	45	51	49	100%	100%	YES	YES	18	15	
1.8	39	39	34	34	47	45	51	49	100%	100%	YES	YES	18	15	
1.9	39	39	35	36	47	45	50	48	100%	100%	YES	YES	18	15	
1.10	40	39	34	34	49	48	50	51	100%	100%	YES	YES	20	15	
1.11	43	43	38	39	65	65	67	67	100%	100%	n/a	n/a	36	16	
1.12	43	43	38	39	65	65	67	67	100%	100%	YES	YES	36	16	
1.13	43	43	39	39	65	65	67	67	100%	100%	n/a	n/a	36	16	
1.14	43	43	39	39	65	65	67	67	100%	100%	YES	YES	36	16	
1.15	43	43	38	38	65	65	67	67	100%	100%	n/a	n/a	36	15	
1.16	43	43	38	38	65	65	67	67	100%	100%	YES	YES	36	15	
1.17	42	44	38	38	65	65	67	67	100%	100%	n/a	n/a	36	15	
1.18	42	44	38	38	65	65	67	67	100%	100%	YES	YES	36	15	
1.19	42	43	38	38	65	65	67	67	100%	100%	n/a	n/a	36	15	
1.20	42	43	38	38	65	65	67	67	100%	100%	YES	YES	36	15	

9 Building 304 Block M- Level 1

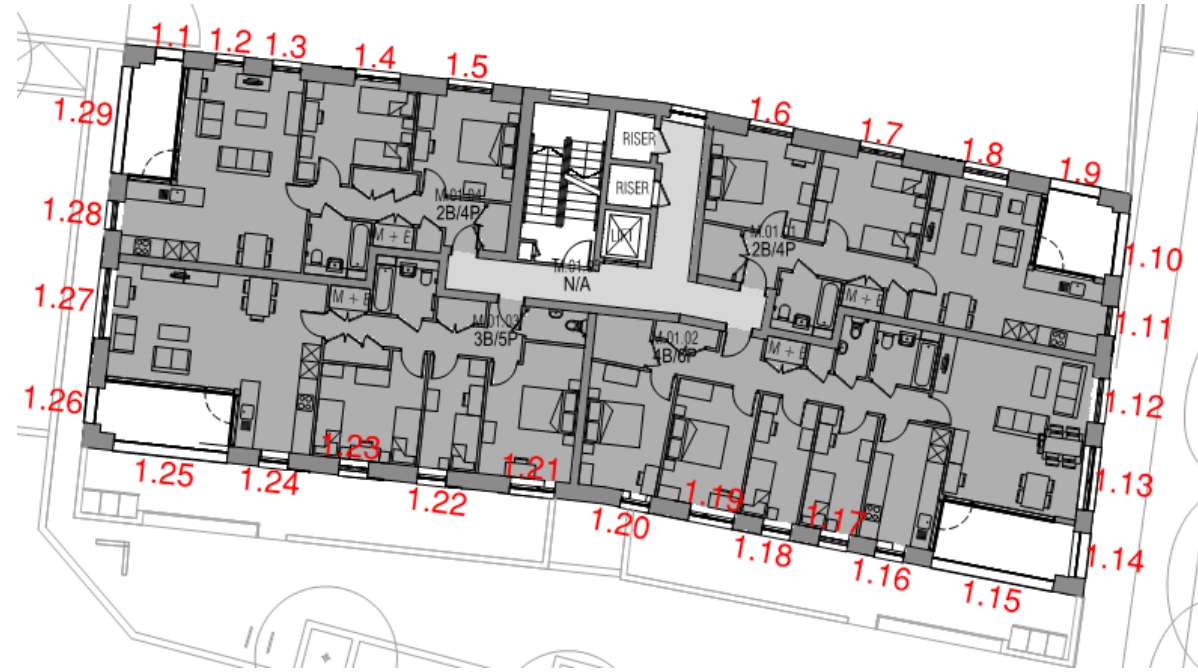


Figure 8 Building 304 Block M– Level 1 floor plan

Table 10 Building 304 Block M – Level 1 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area	Window/ door openable	Window/door insulation (dB)		Notes		
	Wharf & dredgers (L_{Aeq})		Dredgers (L_{A63Hz})		All sources night (L_{Aeq})		All sources day (L_{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP							
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	D_w+C_{tr}	D_{63Hz}	
1.1	44	44	36	36	49	49	54	54	100%	100%	YES	YES	20	15	
1.2	44	44	37	37	48	49	54	54	100%	100%	YES	YES	20	15	
1.3	44	44	38	38	49	50	55	55	100%	100%	YES	YES	21	15	
1.4	44	42	35	35	50	54	56	57	100%	100%	YES	YES	25	15	
1.5	44	42	35	35	50	54	56	57	100%	100%	YES	YES	25	15	
1.6	43	42	36	36	58	58	59	59	100%	100%	YES	YES	29	15	
1.7	42	42	37	37	59	59	56	61	100%	100%	YES	YES	30	15	
1.8	42	43	38	38	61	61	62	63	100%	100%	YES	YES	32	17	
1.9	42	41	37	37	62	61	64	63	100%	100%	YES	YES	33	15	
1.10	42	41	37	37	65	65	67	67	100%	100%	YES	YES	36	15	
1.11	42	42	37	37	65	65	67	67	100%	100%	YES	YES	36	15	
1.12	42	42	37	37	65	65	67	67	100%	100%	YES	YES	36	15	
1.13	42	43	37	37	65	65	67	67	100%	100%	YES	YES	36	15	
1.14	43	43	36	36	65	65	67	67	100%	100%	YES	YES	36	15	
1.15	43	41	30	36	63	63	66	65	100%	100%	YES	YES	34	15	
1.16	42	40	30	37	61	62	65	65	100%	100%	YES	YES	33	15	
1.17	42	41	30	37	59	61	65	64	100%	100%	YES	YES	32	15	
1.18	42	41	30	37	59	59	65	63	100%	100%	YES	YES	31	15	
1.19	42	41	30	37	59	59	64	63	100%	100%	YES	YES	30	15	
1.20	42	41	30	37	59	59	64	62	100%	100%	YES	YES	30	15	
1.21	42	41	30	36	58	59	64	62	100%	100%	YES	YES	30	15	
1.22	42	41	30	35	58	58	64	61	100%	100%	YES	YES	30	15	
1.23	42	41	29	33	58	56	63	61	100%	100%	YES	YES	29	15	
1.24	42	41	29	33	56	56	63	61	100%	100%	YES	YES	27	15	
1.25	42	42	29	30	55	56	63	61	100%	100%	YES	YES	27	15	
1.26	43	41	29	29	53	53	60	53	100%	100%	YES	YES	24	15	
1.27	43	42	29	30	51	51	56	56	100%	100%	YES	YES	22	15	
1.28	41	41	29	30	50	50	58	53	100%	100%	YES	YES	21	15	
1.29	42	39	33	33	50	50	57	52	100%	100%	YES	YES	21	15	

10 Building 304 Block J– Level 2



Figure 9 Building 304 Block J– Level 2 floor plan

Table 11 Building 304– Level 2 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
2.1	42	42	35	35	60	60	61	61	100%	100%	YES	YES	31	15	
2.2	42	42	35	35	60	60	61	61	100%	100%	YES	YES	31	15	
2.3	42	42	35	35	60	60	61	61	100%	100%	YES	YES	31	15	
2.4	43	43	39	39	61	61	62	61	100%	100%	YES	YES	32	16	
2.5	42	43	38	38	62	62	63	63	100%	100%	YES	YES	32	15	
2.6	44	43	36	36	64	64	65	65	100%	100%	YES	YES	35	15	
2.7	44	43	37	37	64	64	65	65	100%	100%	YES	YES	35	15	
2.8	44	43	38	38	64	64	65	65	100%	100%	YES	YES	35	15	
2.9	44	44	38	38	64	64	65	65	100%	100%	YES	YES	35	15	
2.10	41	40	29	35	61	61	64	64	100%	100%	YES	YES	32	12	
2.11	39	40	30	30	58	58	63	63	100%	100%	YES	YES	29	15	
2.12	40	40	30	30	57	57	61	61	100%	100%	YES	YES	28	15	
2.13	40	40	30	30	55	55	59	58	100%	100%	YES	YES	26	15	
2.14	41	41	30	31	55	54	58	57	100%	100%	YES	YES	26	15	
2.15	41	41	30	31	51	50	57	57	100%	100%	YES	YES	22	15	
2.16	41	41	30	31	51	49	57	56	100%	100%	YES	YES	22	15	
2.17	41	41	30	31	50	49	57	55	100%	100%	YES	YES	21	15	
2.18	42	42	30	31	50	48	56	54	100%	100%	YES	YES	21	15	
2.19	41	41	30	31	53	53	55	54	100%	100%	YES	YES	24	15	
2.20	40	40	30	30	53	53	55	54	100%	100%	YES	YES	24	15	
2.21	40	40	30	30	55	54	56	55	100%	100%	YES	YES	26	15	
2.22	40	40	30	30	55	55	57	56	100%	100%	YES	YES	26	15	
2.23	40	40	32	32	56	56	59	58	100%	100%	YES	YES	27	15	

11 Building 304 Block L– Level 2

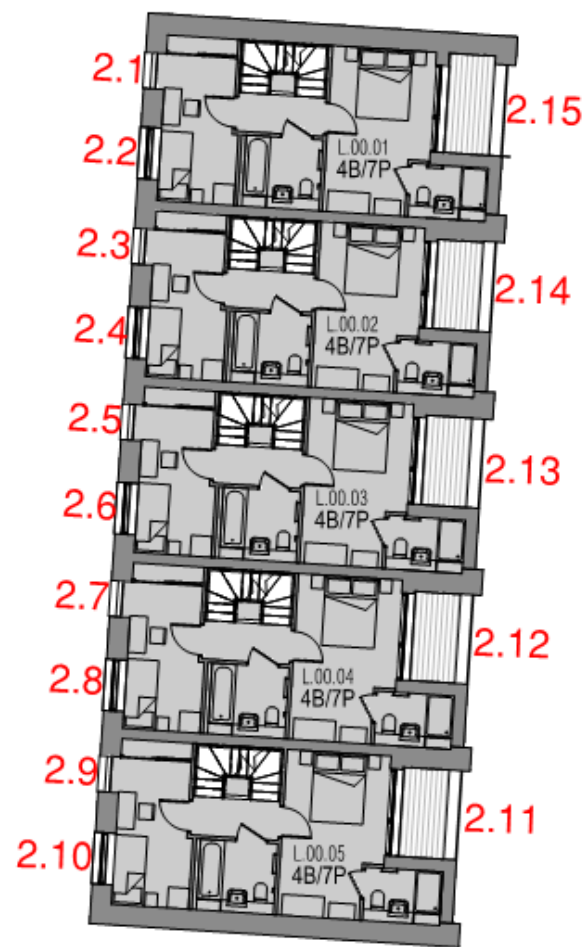


Figure 10 Building 304 Block L– Level 2 floor plan

Table 12 Building Block L – Level 2 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
2.1	39	39	31	31	49	48	54	52	100%	100%	YES	YES	20	15	
2.2	39	39	33	33	48	46	54	49	100%	100%	YES	YES	20	15	
2.3	39	39	33	33	47	45	54	48	100%	100%	YES	YES	20	15	
2.4	39	39	33	33	47	45	53	48	100%	100%	YES	YES	19	15	
2.5	39	40	33	33	47	45	53	48	100%	100%	YES	YES	19	15	
2.6	39	39	33	33	47	46	53	48	100%	100%	YES	YES	19	15	
2.7	39	40	34	35	47	46	53	49	100%	100%	YES	YES	19	15	
2.8	39	40	34	35	47	46	52	49	100%	100%	YES	YES	18	15	
2.9	41	41	36	36	48	46	52	50	100%	100%	YES	YES	19	15	
2.10	42	42	36	36	48	46	54	53	100%	100%	YES	YES	20	15	
2.11	44	44	39	39	64	64	66	66	100%	100%	YES	YES	35	16	
2.12	44	45	39	39	64	64	66	66	100%	100%	YES	YES	35	16	
2.13	44	44	38	38	64	64	66	66	100%	100%	YES	YES	35	15	
2.14	44	44	38	38	64	64	66	66	100%	100%	YES	YES	35	15	
2.15	44	44	39	39	64	64	66	66	100%	100%	YES	YES	35	16	

12 Building 304 Block M- Level 2

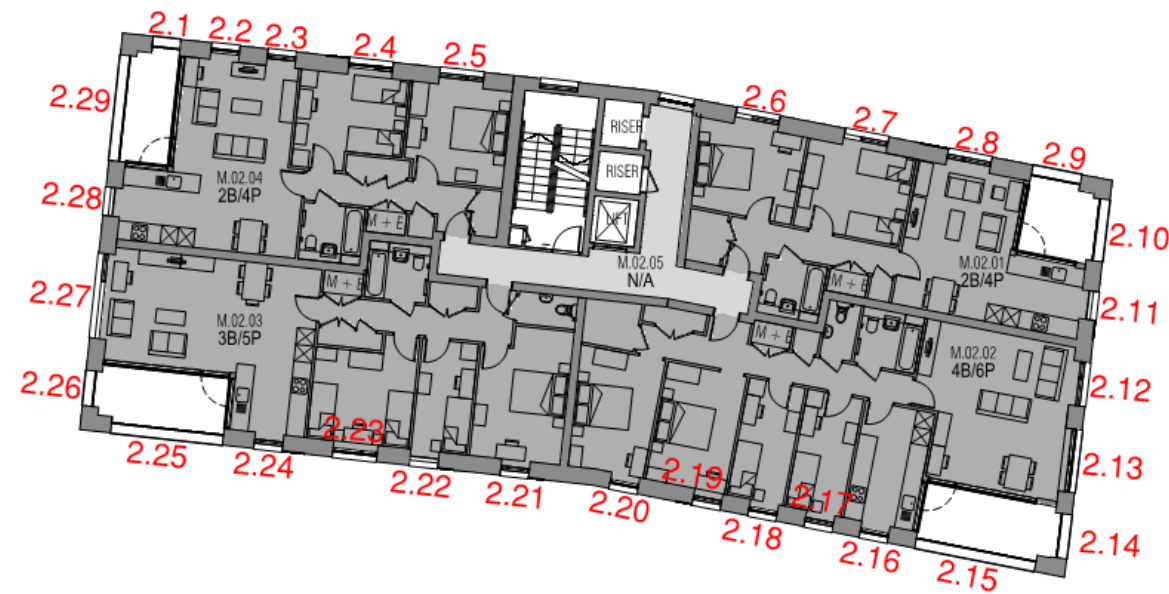


Figure 11 Building 304 Block M– Level 2 floor plan

Table 13 Building 304 Block M- Level 2 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (L_{Aeq})		Dredgers (L_{A63Hz})		All sources night (L_{Aeq})		All sources day (L_{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	D_w+C_{tr}	D_{63Hz}	
2.1	45	45	39	39	50	49	55	54	85%	85%	YES	YES	21	16	Seal and ventilate via balcony or provide Juliet screening arrangement
2.2	45	45	39	39	50	49	56	55	85%	85%	YES	YES	21	16	
2.3	45	45	39	39	50	49	56	55	85%	85%	YES	YES	21	16	
2.4	44	44	38	38	50	50	56	55	100%	100%	YES	YES	22	15	
2.5	44	44	38	38	55	51	57	56	100%	100%	YES	YES	26	15	
2.6	43	43	39	39	58	61	60	60	100%	100%	YES	YES	32	16	
2.7	44	44	39	39	58	59	61	62	100%	100%	YES	YES	30	16	

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (L_{Aeq})		Dredgers (L_{A63Hz})		All sources night (L_{Aeq})		All sources day (L_{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	D_w+C_{tr}	D_{63Hz}	
2.8	44	44	40	40	59	59	62	62	85%	85%	YES	YES	30	17	Seal and ventilate via balcony or provide Juliet screening arrangement
2.9	42	42	39	39	62	58	63	63	100%	100%	YES	YES	33	16	
2.10	42	42	37	37	64	64	66	66	100%	100%	YES	YES	35	14	
2.11	43	43	38	37	65	64	66	66	100%	100%	YES	YES	36	15	
2.12	43	43	38	38	65	64	66	66	100%	100%	YES	YES	36	15	
2.13	43	43	38	38	65	64	66	66	100%	100%	YES	YES	36	15	
2.14	43	43	39	39	64	63	66	66	100%	100%	YES	YES	35	16	
2.15	42	40	30	37	61	60	65	64	100%	100%	YES	YES	32	14	
2.16	42	41	30	36	61	59	64	63	100%	100%	YES	YES	32	13	
2.17	42	41	30	38	61	58	64	63	100%	100%	YES	YES	32	15	
2.18	42	42	30	38	60	57	64	62	100%	100%	YES	YES	31	15	
2.19	42	42	30	38	58	56	64	62	100%	100%	YES	YES	30	15	
2.20	42	42	30	38	57	54	64	62	100%	100%	YES	YES	30	15	
2.21	42	42	30	38	57	54	64	61	100%	100%	YES	YES	30	15	
2.22	42	42	30	36	57	53	64	61	100%	100%	YES	YES	30	15	
2.23	42	42	29	34	56	53	64	61	100%	100%	YES	YES	30	15	
2.24	42	42	29	34	56	53	64	61	100%	100%	YES	YES	27	15	
2.25	42	41	29	30	56	52	64	61	100%	100%	YES	YES	27	15	
2.26	43	40	29	30	53	47	60	56	100%	100%	YES	YES	24	15	
2.27	43	41	29	30	52	47	58	53	100%	100%	YES	YES	23	15	
2.28	43	42	29	30	51	47	57	51	100%	100%	YES	YES	22	15	
2.29	43	43	36	37	51	46	57	53	100%	100%	YES	YES	22	15	

13 Building 304 Block J– Level 3



Figure 12 Building 304 Block J– Level 3 floor plan

Table 14 Building 304 Block J – Level 3 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
3.1	43	43	36	36	60	59	61	61	100%	100%	YES	YES	30	15	Window can be openable taking 1 dB low frequency allowance
3.2	43	43	36	36	60	58	61	61	100%	100%	YES	YES	30	15	
3.3	43	43	36	36	60	58	61	61	100%	100%	YES	YES	30	15	
3.4	43	43	36	36	60	59	61	61	100%	100%	YES	YES	30	16	
3.5	44	44	40	40	60	60	61	62	0%	0%	NO	NO	31	17	
3.6	44	44	40	40	61	61	63	61	85%	85%	YES	YES	32	17	Seal and ventilate via balcony or provide Juliet screening arrangement
3.7	44	44	40	40	63	63	65	65	85%	85%	YES	YES	34	15	
3.8	45	45	40	40	63	63	66	65	85%	85%	YES	YES	34	16	Juliet screening arrangement
3.9	45	45	40	40	63	63	65	65	85%	85%	YES	YES	34	16	Seal and ventilate via balcony or provide Juliet screening arrangement
3.10	42	42	31	31	61	60	63	63	100%	100%	YES	YES	32	15	
3.11.	41	41	31	31	59	57	62	62	100%	100%	YES	YES	30	15	
3.12	41	41	31	31	58	56	61	61	100%	100%	YES	YES	29	15	
3.13	42	42	31	31	56	55	59	59	100%	100%	YES	YES	27	15	
3.14	42	42	31	31	55	54	59	57	100%	100%	YES	YES	26	15	
3.15	42	42	31	31	52	50	58	56	100%	100%	YES	YES	24	15	
3.16	42	42	31	31	51	49	58	55	100%	100%	YES	YES	24	15	
3.17	43	43	31	31	51	49	57	55	100%	100%	YES	YES	23	16	
3.18	43	43	31	31	50	49	57	54	100%	100%	YES	YES	21	16	
3.19	41	41	31	31	53	52	55	54	100%	100%	YES	YES	24	15	
3.20	40	41	30	31	53	53	56	54	100%	100%	YES	YES	24	15	
3.21	40	41	30	31	54	54	56	54	100%	100%	YES	YES	25	15	
3.22	41	41	30	31	55	54	57	56	100%	100%	YES	YES	26	15	
3.23	40	41	30	31	56	55	59	57	100%	100%	YES	YES	27	15	

14 Building 304 Block L– Level 3

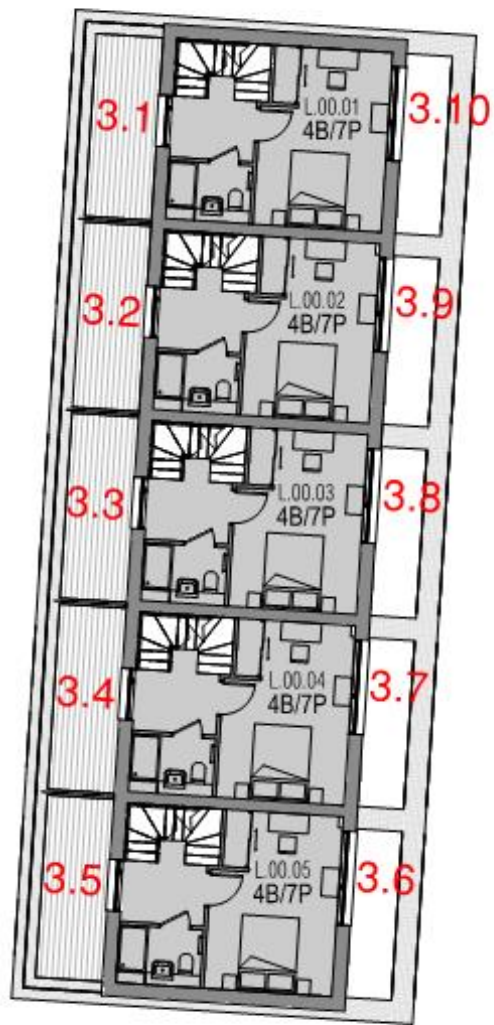


Figure 13 Building 304 Block L– Level 3 floor plan

Table 15 Building 304 – Level 3 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
3.1	42	42	36	36	51	50	56	54	100%	100%	YES	YES	17	15	
3.2	42	43	38	38	50	48	54	51	100%	100%	YES	YES	16	15	
3.3	43	43	38	38	50	48	54	51	100%	100%	YES	YES	16	15	
3.4	43	43	39	39	50	48	54	51	100%	100%	YES	YES	16	15	
3.5	43	43	39	39	50	49	65	51	100%	100%	YES	YES	21	15	
3.6	43	43	38	38	62	62	65	65	100%	100%	YES	YES	33	16	
3.7	43	43	38	38	62	62	66	66	100%	100%	YES	YES	33	16	
3.8	43	43	39	39	62	62	66	66	100%	100%	YES	YES	33	16	
3.9	43	43	37	37	62	62	66	66	100%	100%	YES	YES	33	16	
3.10	43	43	37	37	62	62	66	66	100%	100%	YES	YES	33	16	

15 Building 304 Block M– Level 3



Figure 14 Building 304 Block M– Level 3 floor plan

Table 16 Building 304 Block M – Level 3 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
3.1	45	45	39	39	50	49	55	55	85%	85%	YES	YES	21	16	Seal and ventilate via balcony or provide Juliet screening arrangement
3.2	45	45	39	39	50	50	55	55	85%	85%	YES	YES	21	16	
3.3	45	45	39	39	50	50	55	55	85%	85%	YES	YES	21	16	Seal and ventilate via balcony or provide Juliet screening arrangement
3.4	45	45	39	39	51	51	56	56	85%	85%	YES	YES	22	16	Juliet screening arrangement
3.5	44	45	39	39	51	54	57	56	100%	85%	YES	YES	25	16	Juliet screening arrangement
3.6	45	45	39	39	57	57	60	60	85%	85%	YES	YES	28	16	Juliet screening arrangement
3.7	45	45	40	40	58	58	61	61	85%	85%	YES	YES	29	17	Juliet screening arrangement
3.8	45	45	40	40	60	59	62	62	85%	85%	YES	YES	31	17	Juliet screening arrangement

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
3.9	44	43	40	40	60	60	63	62	85%	85%	YES	YES	31	17	Seal and ventilate via balcony or provide Juliet screening arrangement
3.10	44	44	37	37	63	63	66	66	100%	100%	YES	YES	34	15	
3.11	44	44	39	39	63	63	66	66	100%	100%	YES	YES	34	16	
3.12	44	44	39	39	63	63	66	66	100%	100%	YES	YES	34	16	Seal and ventilate via balcony or provide Juliet screening arrangement
3.13	44	44	39	39	63	63	66	66	100%	100%	YES	YES	34	16	Seal and ventilate via balcony or provide Juliet screening arrangement
3.14	44	44	39	39	62	62	66	65	100%	100%	YES	YES	33	16	Seal and ventilate via balcony or provide Juliet screening arrangement
3.15	42	41	31	37	61	60	65	64	100%	100%	YES	YES	32	15	
3.16	42	41	30	38	59	58	64	62	100%	100%	YES	YES	30	15	
3.17	42	42	30	38	59	58	64	62	100%	100%	YES	YES	30	15	
3.18	42	42	30	38	59	57	64	61	100%	100%	YES	YES	30	15	
3.19	42	42	30	39	58	55	64	61	100%	100%	YES	YES	30	16	
3.20	42	42	30	39	57	54	64	61	100%	100%	YES	YES	30	16	
3.21	42	43	30	39	57	54	64	61	100%	100%	YES	YES	30	16	
3.22	42	42	30	39	57	53	64	61	100%	100%	YES	YES	30	16	
3.23	42	43	30	38	57	53	64	61	100%	100%	YES	YES	30	15	
3.24	43	42	30	38	57	53	64	61	100%	100%	YES	YES	30	15	Seal and ventilate via balcony or provide Juliet screening arrangement
3.25	43	43	30	30	56	52	64	61	100%	100%	YES	YES	27	15	
3.26	43	43	30	30	53	48	60	57	100%	100%	YES	YES	24	15	
3.27	43	43	30	30	52	47	58	63	100%	100%	YES	YES	24	15	
3.28	43	41	30	30	52	47	58	54	100%	100%	YES	YES	23	15	
3.29	44	40	37	37	52	45	57	53	100%	100%	YES	YES	23	15	

16 Building 304 Block J– Level 4

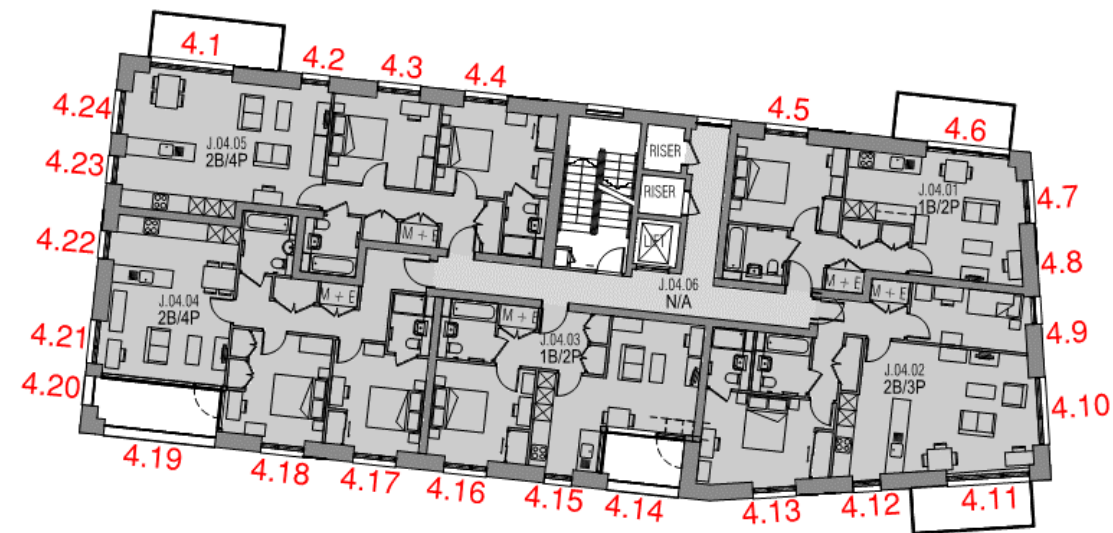


Figure 15 Building 304 Block J– Level 3 floor plan

Table 17 Building 304 – Level 4 floor plan predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
4.1	44	44	37	37	58	58	61	60	100%	100%	YES	YES	29	15	Juliet screening arrangement
4.2	44	44	37	37	59	59	61	60	100%	100%	YES	YES	30	15	
4.3	44	44	37	37	59	59	61	60	100%	100%	YES	YES	30	15	
4.4	44	44	37	37	59	59	61	60	100%	100%	YES	YES	30	15	
4.5	44	45	40	40	60	60	61	61	85%	85%	YES	YES	31	17	
4.6	44	45	41	41	60	60	62	62	70%	70%	YES	YES	31	18	
4.7	45	45	39	39	62	62	65	65	85%	85%	YES	YES	33	16	
4.8	45	45	41	41	62	62	65	65	70%	70%	YES	YES	33	18	
4.9	45	45	41	41	62	62	65	65	70%	70%	YES	YES	33	18	
4.10	45	45	40	41	62	62	65	65	85%	70%	YES	YES	33	18	
4.11.	42	42	31	31	60	60	63	63	100%	100%	YES	YES	31	15	Seal and ventilate via balcony or provide Juliet screening arrangement
4.12	42	42	32	32	59	59	61	62	100%	100%	YES	YES	30	15	
4.13	42	42	31	31	57	58	60	60	100%	100%	YES	YES	29	15	
4.14	42	42	32	32	55	56	59	59	100%	100%	YES	YES	27	15	
4.15	42	42	32	32	54	53	59	58	100%	100%	YES	YES	25	15	
4.16	41	41	32	32	52	50	58	57	100%	100%	YES	YES	24	15	
4.17	41	41	32	32	51	50	58	56	100%	100%	YES	YES	24	15	
4.18	41	41	31	31	51	49	58	55	100%	100%	YES	YES	24	15	
4.19	41	41	32	32	51	49	57	54	100%	100%	YES	YES	22	15	
4.20	41	43	33	33	52	51	56	54	100%	100%	YES	YES	23	15	
4.21	41	41	32	32	53	52	56	54	100%	100%	YES	YES	24	15	
4.22	42	42	32	32	54	53	56	54	100%	100%	YES	YES	25	15	
4.23	41	42	32	32	55	54	57	55	100%	100%	YES	YES	26	15	
4.24	41	42	33	33	56	55	58	57	100%	100%	YES	YES	27	15	

17 Building 304 Block M– Level 4



Figure 16 Building 304 Block M– Level 4 floor plan

Table 18 Building 304 Block M – Level 4 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
4.1	46	46	41	41	51	50	56	55	70%	70%	YES	YES	22	18	Seal and ventilate via balcony or provide Juliet screening arrangement
4.2	46	46	41	41	51	50	56	55	70%	70%	YES	YES	22	18	
4.3	46	46	41	41	51	50	56	55	70%	70%	YES	YES	22	18	Seal and ventilate via balcony or provide Juliet screening arrangement
4.4	46	46	41	41	51	51	57	55	70%	70%	YES	YES	23	18	Juliet screening arrangement
4.5	46	46	41	41	52	57	58	56	70%	70%	YES	YES	28	18	Juliet screening arrangement
4.6	46	46	41	41	57	58	60	60	70%	70%	YES	YES	29	18	Juliet screening arrangement
4.7	45	45	41	41	58	58	60	61	70%	70%	YES	YES	29	18	Juliet screening arrangement

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
4.8	45	45	41	41	59	59	62	61	70%	70%	YES	YES	30	18	Juliet screening arrangement
4.9	44	44	40	40	60	60	62	62	85%	85%	YES	YES	31	17	
4.10	45	45	38	38	62	62	65	65	85%	85%	YES	YES	33	15	
4.11	45	45	39	39	63	63	65	65	85%	85%	YES	YES	34	16	
4.12	45	45	40	40	63	63	66	65	85%	85%	YES	YES	34	17	Seal and ventilate via balcony or provide Juliet screening arrangement
4.13	45	45	40	40	63	62	66	65	85%	85%	YES	YES	34	17	Seal and ventilate via balcony or provide Juliet screening arrangement
4.14	45	45	40	40	62	62	65	65	85%	85%	YES	YES	33	17	*Window can be openable taking 1 dB low frequency allowance
4.15	43	42	32	38	60	60	65	64	100%	100%	YES	YES	31	15	
4.16	42	42	31	38	59	58	64	62	100%	100%	YES	YES	30	15	
4.17	42	42	31	38	59	58	64	62	100%	100%	YES	YES	30	15	
4.18	42	44	31	40	58	57	64	62	100%	0%*	NO*	YES	30	17	
4.19	42	44	31	40	57	56	64	62	100%	0%*	NO*	YES	30	17	
4.20	42	44	31	40	57	54	64	61	100%	0%*	NO*	YES	30	17	
4.21	42	44	30	41	57	54	64	61	100%	0%*	NO*	YES	30	18	
4.22	43	43	30	41	57	54	64	61	100%	0%*	NO*	YES	30	18	
4.23	43	44	30	38	57	54	64	61	100%	100%	YES	YES	30	15	
4.24	43	43	30	38	57	53	64	61	100%	100%	YES	YES	28	15	
4.25	43	44	30	38	56	52	64	61	100%	100%	YES	YES	27	15	
4.26	43	42	31	32	53	49	60	57	100%	100%	YES	YES	24	15	
4.27	44	43	31	32	53	48	64	63	100%	100%	YES	YES	25	15	
4.28	43	43	31	33	52	48	58	54	100%	100%	YES	YES	23	15	
4.29	45	45	39	40	52	47	59	53	85%	85%	YES	YES	23	17	

18 Building 304 Block J– Level 5



Figure 17 Building 304 Block J– Level 5 floor plan

Table 19 Building 304 Block J – Level 5 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
5.1	45	46	41	41	58	58	60	60	70%	70%	YES	YES	29	18	Seal and ventilate via balcony or provide Juliet screening arrangement
5.2	45	46	41	41	58	58	60	60	70%	70%	YES	YES	29	18	
5.3	45	46	40	41	58	58	61	60	85%	70%	YES	YES	29	18	
5.4	45	46	40	40	58	58	61	60	85%	70%	YES	YES	29	17	Juliet screening arrangement
5.5	45	45	41	42	59	59	61	60	70%	45%	YES	YES	30	19	Juliet screening arrangement. 70% open area with 1 dB low frequency allowance
5.6	45	45	42	42	59	59	61	61	45%	45%	YES	YES	30	19	70% open area with 1 dB low frequency allowance
5.7	46	46	41	41	61	61	64	64	70%	70%	YES	YES	32	18	
5.8	46	46	41	41	62	61	64	64	70%	70%	YES	YES	33	18	
5.9	43	43	33	33	59	59	62	62	100%	100%	YES	YES	30	15	
5.10	43	43	33	34	58	57	61	60	100%	100%	YES	YES	29	15	
5.11	43	43	33	33	57	55	59	58	100%	100%	YES	YES	28	15	
5.12	42	42	33	33	53	51	59	58	100%	100%	YES	YES	24	15	
5.13.	41	42	33	34	52	51	58	57	100%	100%	YES	YES	24	15	
5.14	41	42	33	34	51	49	58	56	100%	100%	YES	YES	24	15	
5.15	42	42	34	34	51	49	58	55	100%	100%	YES	YES	24	15	
5.16	42	42	34	34	51	49	57	55	100%	100%	YES	YES	22	15	
5.17	43	43	34	34	53	52	56	54	100%	100%	YES	YES	24	15	
5.18	44	44	34	34	54	53	56	54	100%	100%	YES	YES	25	15	
5.19	44	44	34	34	54	54	56	55	100%	100%	YES	YES	25	15	
5.20	44	44	34	34	55	54	57	56	100%	100%	YES	YES	26	15	

19 Building 304 Block M– Level 5

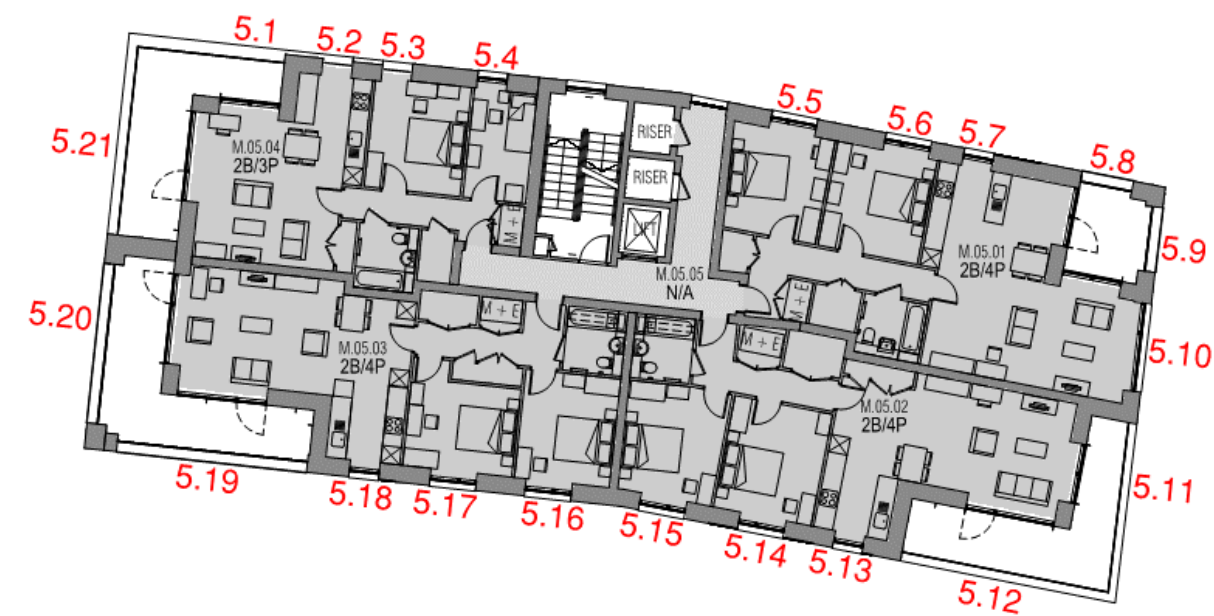


Figure 18 Building 304 Block M – Level 5 floor plan

Table 20 Building 304 Block M – Level 5 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
5.1	47	47	42	42	52	51	57	55	45%	45%	YES	YES	23	19	Fix window to avoid excess balcony screening. Ventilate via 5.21.
5.2	46	46	42	42	51	50	57	55	45%	45%	YES	YES	23	19	Seal and ventilate via balcony or provide Juliet screening arrangement. 70% open area with 1 dB low frequency allowance
5.3	46	46	42	42	51	50	57	55	45%	45%	YES	YES	23	19	Seal and ventilate via balcony or provide Juliet screening arrangement. 70% open area with 1 dB low frequency allowance

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
5.4	46	46	42	42	52	51	58	56	45%	45%	YES	YES	24	19	Juliet screening arrangement. 70% open area with 1 dB low frequency allowance
5.5	46	46	41	41	57	57	60	59	70%	70%	YES	YES	29	18	Juliet screening arrangement
5.6	45	46	41	41	58	58	61	61	70%	70%	YES	YES	30	18	Juliet screening arrangement
5.7	45	46	41	41	58	58	61	61	70%	70%	YES	YES	30	18	Seal and ventilate via balcony or provide Juliet screening arrangement.
5.8	46	46	41	41	59	59	62	62	70%	70%	YES	YES	33	18	
5.9	46	46	40	40	62	62	65	65	70%	70%	YES	YES	33	18	
5.10	46	46	40	40	62	62	65	65	70%	70%	YES	YES	31	16	Seal and ventilate via balcony or provide Juliet screening arrangement.
5.11	46	46	40	40	62	62	65	65	70%	70%	YES	YES	31	16	
5.12	43	44	33	39	60	62	65	64	100%	100%	YES	YES	30	18	
5.13	43	44	33	39	60	59	64	64	100%	100%	YES	YES	30	18	
5.14	43	46	33	39	59	58	64	63	100%	70%	YES	YES	30	18	Juliet screening arrangement
5.15	43	46	33	41	59	57	64	62	100%	70%	YES	YES	30	19	Juliet screening arrangement. 70% open area with 1 dB low frequency allowance
5.16	43	46	33	41	58	55	64	62	100%	70%	YES	YES	30	19	Juliet screening arrangement. 70% open area with 1 dB low frequency allowance
5.17	43	46	33	42	57	55	64	62	100%	45%	YES	YES	30	16	Juliet screening arrangement. 70% open area with 1 dB low frequency allowance
5.18	43	45	34	39	57	54	66	62	100%	85%	YES	YES	25	15	Juliet screening arrangement
5.19	44	46	34	39	57	51	64	61	100%	70%	YES	YES	24	15	Seal and ventilate via balcony or provide Juliet screening arrangement.
5.20	46	45	34	34	54	49	61	57	70%	85%	YES	YES	23	15	
5.21	45	46	40	40	52	49	59	53	85%	70%	YES	YES	23	17	